Hospital-Level Variation in Gastric Cancer Treatment and Its Effects on Mortality and Length of Stay in the Chilean Public Health System

Vicente Rodríguez,[a,*] José Tomás Amat,[a] Sebastián Herrera[a]

[a]*School of Engineering, Universidad del Desarrollo, Santiago, Chile.* [*]*vjrodrig2004@gmail.com*

Background Gastric cancer is the leading cause of cancer-related mortality in the Chilean public health system, yet the extent to which hospital characteristics—rather than patient clinical profiles—determine outcomes remains unquantified. We investigated whether the treating hospital explains significant variation in length of stay, number of procedures, and in-hospital mortality in gastric cancer patients after controlling for clinical severity.

Methods We performed a cross-sectional analysis of Chile's national DRG database (2019–2024), identifying 17,335 gastric cancer discharges (C16.*) and restricting regression analyses to 9,855 discharges across 15 high-volume public hospitals. We applied Kruskal-Wallis with Dunn-Bonferroni post-hoc (H■), multivariable logistic regression (H■), and OLS with HC3 robust errors (H■). A random-intercept mixed model estimated the intraclass correlation coefficient (ICC). Socioeconomic context was incorporated via linkage with 2024 Municipal HDI quintiles, and hospital typology was characterized by K-means clustering.

Findings Procedure distribution differed significantly across hospitals (Kruskal-Wallis $H = 412.3$, $p<0.001$, $\varepsilon^2 = 0.080$); 61.8% of hospital pairs showed significant pairwise differences after Bonferroni correction. Each additional procedure increased in-hospital mortality odds by 5% (OR = 1.05; 95% CI 1.02–1.08; $p<0.001$), with GRD severity as the dominant predictor (OR = 6.12). Each procedure was also associated with a 9.7% increase in length of stay ($\beta = 0.093$; $R^2 = 0.636$). The ICC indicated that 18.4% of variance in log-length of stay is attributable to the hospital rather than to patient characteristics. Patients from the lowest HDI quintile showed 2.8 percentage-points higher in-hospital mortality than those from the highest quintile ($p = 0.003$). K-means identified three distinct hospital typologies.

Interpretation Hospital-level variation in gastric cancer care in Chile is structural, statistically robust, and socioeconomically patterned. The treating hospital explains nearly one-fifth of the variance in length of stay after controlling for patient characteristics. These findings support policies targeting institutional protocol standardization and differential resource allocation to territorially disadvantaged areas.

Keywords: gastric cancer · hospital variation · intraclass correlation · Chile · DRG database · health equity · socioeconomic determinants

Research in context

Evidence before this study

Hospital-level variation in oncological outcomes has been documented in high-income countries (Kamaraju et al., 2022; OECD, 2023). Evidence from Latin America using national DRG data is scarce. Prior work on Chile's database examined multimorbidity (Bernal et al., 2026) but not inter-hospital variability or its socioeconomic determinants.

Added value of this study

First nationwide analysis of hospital-level variation in gastric cancer in Chile (2019–2024, 15 high-volume hospitals). We quantify the institutional contribution through a random-intercept ICC of 18.4%, link variation to municipal HDI quintiles, and characterize three distinct hospital typologies via K-means clustering—all providing actionable evidence for health system governance.

Implications of all available evidence

The treating hospital explains nearly one fifth of LOS variance after conditioning on patient clinical characteristics. Patients from the lowest-HDI quintile face 2.8 pp higher in-hospital mortality. Protocol standardization and differential resource allocation to socioeconomically disadvantaged territories are the priority policy responses.

INTRODUCTION

Cancer is the second leading cause of death in Chile and the primary contributor to years of life lost prematurely.[1] Among digestive cancers, gastric cancer (ICD-10: C16.*) ranks first in incidence and mortality within the public health system, with five-year survival rates substantially lower than those in high-income countries.[2]

A growing body of evidence shows that hospital-level variation in cancer care is not fully explained by patient clinical characteristics.[3,4] This *unjustified clinical variability* (Wennberg, 2010)[5] has direct implications for health equity: when outcomes are partly determined by the institution of care rather than by clinical need, access to quality treatment becomes a function of geography and socioeconomic status rather than medical necessity.

In Chile, the public health system (FONASA) covers approximately 80% of the population through a network of hospitals with heterogeneous complexity, resource endowment, and geographic distribution. Hospitals serving lower-income territories—identified through the Municipal Human Development Index (HDI)—face compounded barriers: higher proportions of late-stage presentations, emergency admissions, and constrained capacity.

Despite the availability of a nationwide DRG database, no study has quantified inter-hospital variation in gastric cancer outcomes using multilevel methods, or linked this variation to the socioeconomic profile of the territory of care. We address this gap by testing three pre-specified hypotheses: (H■) procedure distributions differ across hospitals; (H■) procedure count is independently associated with in-hospital mortality; (H■) procedure count independently predicts length of stay. We further quantify the institutional ICC, characterize the socioeconomic gradient, and identify hospital typologies by K-means clustering.

METHODS

Data source and study population

Data were obtained from Chile's national DRG database (MINSAL/FONASA, 2019–2024), covering 100% of hospital discharge records from public institutions. We selected all discharges with a primary diagnosis of gastric cancer (C16.*). Five cleaning stages were applied: (1) restriction to oncological ICD-10 codes C00–D49; (2) exclusion of obstetric admissions; (3) truncation of length-of-stay outliers at the 99th percentile (P■■ = 25 days); (4) restriction to C16.* subcodes; and (5) restriction to the 15 highest-volume hospitals (≥30 C16.* discharges each) for regression analyses.

Variables

Outcomes: (i) procedure count; (ii) in-hospital mortality (binary); (iii) length of stay (days), log-transformed [log(1+LOS)] to correct right skew (skewness 1.63 raw, −0.32 post-transformation). *Clinical covariates:* age; GRD severity (1–4 ordinal); GRD relative weight; comorbidity count (secondary diagnoses, simplified Charlson-Deyo proxy). *Institutional variable:* hospital (fixed or random effect). *Socioeconomic variable:* Municipal HDI 2024, linked by normalized name-matching and categorized into five quintiles.

Statistical analysis

Normality was assessed via Shapiro-Wilk (subsampled n = 5,000). Given rejection (W = 0.922, p<0.001), H■ was tested by Kruskal-Wallis across the top-25 hospitals, followed by Dunn-Bonferroni post-hoc comparisons. Effect size: ε^2. H■ used multivariable logistic regression [logit(mortality) = β■ + β■·proc + β■·age + β■·sev + β■·wt + β■·comorb + γC(hospital)] estimated via BFGS; discrimination assessed by AUC-ROC on a stratified 75/25 split. H■ used OLS with HC3 robust standard errors and back-transformation via np.expm1. A random-intercept MixedLM (REML) estimated the ICC. Socioeconomic analyses used Kruskal-Wallis (LOS) and chi-squared (mortality) across HDI quintiles. K-means clustering (k = 3, silhouette-selected) operated on five standardized hospital-level indicators. All analyses in Python 3.13; $\alpha = 0.05$.

RESULTS

Study population characteristics

After the five-stage pipeline, 17,335 gastric cancer discharges were identified from 421,824 oncological discharges (4.1%). The regression sample comprised 9,855 discharges across 15 hospitals. Mean age was 63.2 ± 13.1 years; 60.3% were male. Median LOS was 7 days (IQR 3–12); median procedures per admission 3 (IQR 1–6). Overall in-hospital mortality was 4.05%. Key characteristics are presented in Table 1.

Characteristic	C00–D49	C16.*
n (discharges)	421,824	9,855†
Age, mean ± SD (yr)	59.8 ± 15.4	63.2 ± 13.1
Male sex, %	48.1	60.3
LOS, median [IQR]	5 [2–10]	7 [3–12]
Procedures, median [IQR]	2 [1–4]	3 [1–6]
GRD severity, mean±SD	1.82 ± 0.91	2.14 ± 0.88
GRD weight, mean±SD	1.31 ± 1.02	1.68 ± 1.24
Comorbidity, mean±SD	1.43 ± 1.78	1.61 ± 1.84
Emergency admission, %	38.9	42.7
In-hospital mortality, %	2.31	4.05

Table 1. Descriptive statistics of the study population. †Restricted to 15 high-volume hospitals. LOS = length of stay; IQR = interquartile range.

H■: Kruskal-Wallis — procedural variability

The Kruskal-Wallis test across 25 hospitals (N = 14,283) yielded $H(24) = 412.3$, $p<0.001$, $\varepsilon^2 = 0.080$ (medium-to-large effect). Dunn-Bonferroni post-hoc identified significant pairwise differences in **61.8% of all hospital pairs** (Fig. 1). This generalized pattern confirms that procedural variability is structural—not driven by outlier institutions—but reflects system-wide heterogeneity in treatment intensity.

H■: Logistic regression — in-hospital mortality

The multivariable logistic model (n = 9,855) achieved AUC-ROC = 0.823. GRD severity dominated prediction (OR = 6.12; Table 2). Each additional procedure increased mortality odds by 5% (OR = 1.05; 95% CI 1.02–1.08; $p<0.001$), though this likely reflects indication confounding. Hospital fixed effects were collectively significant ($\chi^2 = 78.4$, $p<0.001$).

Predictor	OR	95% CI	p
Procedures (per unit)	1.05	1.02–1.08	<0.001
Age (per year)	1.01	0.99–1.02	0.341
GRD severity	6.12	5.42–6.91	<0.001
GRD weight	0.88	0.82–0.95	0.001
Comorbidity (per count)	1.06	1.01–1.11	0.018
Hospital (FE)	—	—	<0.001
Pseudo-R^2 (McFadden)	0.182	—	—
AUC-ROC	0.823	—	—

Table 2. Multivariable logistic regression for in-hospital mortality. Hospital fixed effects included but not shown. OR = odds ratio; CI = confidence interval.

H■: OLS regression — length of stay

The OLS model achieved $R^2 = 0.636$ (adj. $R^2 = 0.635$), $F(19, 9835) = 978.1$, $p<0.001$. Each additional procedure was associated with a **9.7% increase in LOS** ($\beta = 0.093$; $p<0.001$; Table 3). GRD severity had the largest marginal effect (+27.3%). Hold-out evaluation yielded MAE = 3.47 days and RMSE = 5.61 days.

Predictor	β	SE	95% CI	Δ%
Procedures (per unit)	0.093	0.004	0.085–0.101	+9.7
Age (per year)	0.002	0.001	0.001–0.004	+0.2
GRD severity	0.241	0.011	0.220–0.263	+27.3
GRD weight	0.078	0.008	0.062–0.094	+8.1
Comorbidity (per count)	0.031	0.006	0.020–0.043	+3.2
R^2 / Adj. R^2	0.636	/	0.635	—
MAE (test)	3.47 d	RMSE	5.61 d	—

Table 3. OLS regression for log(1+LOS), HC3 robust standard errors. $\Delta\% = (\exp(\beta)-1)\times 100$. Hospital fixed effects included but not shown.

Multilevel model and ICC

The random-intercept MixedLM (REML) estimated $\sigma^2_{hospital} = 0.041$ and $\sigma^2_{residual} = 0.183$, yielding **ICC = 0.184 (18.4%)**. This indicates that approximately one fifth of the total variance in log-LOS is attributable to the treating hospital, not to patient characteristics. Fixed-effect estimates were consistent with OLS results (within 5% for all continuous predictors), confirming robustness to model specification.

Socioeconomic gradient: HDI quintile analysis

HDI linkage succeeded for 87.3% of C16.* discharges. In-hospital mortality followed a monotonic socioeconomic gradient: Q1 (lowest HDI): 5.2%; Q2: 4.7%; Q3: 4.1%; Q4: 3.9%; Q5: 2.4% (Kruskal-Wallis $H(4)=28.7$, $p<0.001$; $\chi^2(4)=14.2$, $p=0.003$). The **Q1–Q5 absolute difference was 2.8 percentage points** (117% relative excess risk). Q1 patients showed longer adjusted LOS (median 8 vs. 6 days) and higher emergency admission rates (50.3% vs. 35.6%), consistent with later-stage presentation.

Hospital typology: K-means clustering (k = 3)

Cluster	n hosp.	Mortality %	LOS (d)	Emerg. %
1 — High-complexity/Urgent	4	6.1	9.3	54
2 — Surgical/Specialized	6	3.8	8.6	38
3 — General/Low-complexity	5	2.9	5.8	41

Table 4. K-means cluster centroids (silhouette = 0.43). Cluster 1 hospitals disproportionately serve Q1–Q2 HDI municipalities.

DISCUSSION

Principal findings

This study provides the first quantification of hospital-level variation in gastric cancer outcomes across Chile's public health network using multilevel methods and linkage to socioeconomic territorial indicators. Our principal findings are: (i) procedural variability is structural and generalized; (ii) procedure count is independently associated with both mortality and LOS; (iii) the treating hospital explains 18.4% of LOS variance; and (iv) patients from the most socioeconomically deprived municipalities experience a 2.8-point excess mortality that cannot be attributed to clinical case-mix alone.

Comparison with international evidence

Our ICC of 18.4% falls within the 15–25% range reported by the OECD for hospital-level contributions to LOS variance.[7] Kamaraju et al.[6] documented up to 8 pp mortality differences across US gastric cancer hospitals after risk adjustment, consistent with our Q1–Q5 differential. The socioeconomic gradient we observe parallels evidence from Colombia[8] and Brazil[9], reinforcing the generalizability of these findings to Latin American middle-income health systems.

Mechanistic interpretation and socioeconomic implications

Two mechanisms likely explain our findings. First, *late-presentation bias*: patients from lower-HDI municipalities arrive with higher GRD severity and more emergency presentations, captured by our clinical covariates. Second, residual institutional variation (the ICC) suggests that differences in treatment protocols, infrastructure, and human resources contribute independently to outcomes. The K-means clustering reinforces this dual mechanism: Cluster 1 hospitals—highest mortality, most emergencies—disproportionately serve Q1–Q2 municipalities, closing a cycle of socioeconomic disadvantage: lower preventive access → late diagnosis → emergency presentation → worse institutional outcomes.

Limitations

The DRG database does not include cancer staging (TNM), histology, ECOG performance status, or specific treatment regimens—variables that would explain part of the residual variance (36% in H■) and could modify the ICC. The hospitalization-episode unit of analysis prevents individual-level survival analysis. The HDI linkage operates at the municipal level, introducing ecological fallacy risk. The observational design precludes causal inference.

Conclusions and policy implications

Hospital-level variation in gastric cancer care in Chile is substantial, statistically robust, and socioeconomically patterned. Three policy implications follow: **(1) Standardize oncological protocols** in Cluster 1 hospitals; **(2) invest differentially** in establishments serving low-HDI municipalities; **(3) implement early referral systems** for complex cases. These recommendations are actionable within the current FONASA governance framework without requiring legislative change—only evidence-based resource allocation and protocol enforcement.

CONTRIBUTORS

V.R., J.T.A., and S.H. contributed equally to conceptualization, data curation, formal analysis, visualization, and writing. V.R. had primary responsibility for statistical modeling and code development.

DECLARATION OF INTERESTS

The authors declare no competing interests. This work was carried out as a final project for *Análisis de Datos e Inferencia Estadística,* School of Engineering, Universidad del Desarrollo, Santiago, Chile (2026).

DATA SHARING

The GRD database is publicly available from Chile's Ministry of Health (datos.gob.cl). Analysis code and outputs: github.com/SebaGRT/Proyecto-Hospitalizacion.

REFERENCES

1. Ministerio de Salud de Chile. *Situación de Salud de Chile 2023.* MINSAL; 2023.
2. Global Burden of Disease Collaborative Network. *GBD Study 2021.* IHME; 2022.
3. Wennberg JE. *Tracking Medicine.* Oxford University Press; 2010.
4. OECD. *Health at a Glance 2023.* OECD Publishing; 2023.
5. Wennberg JE. Unwarranted variations in healthcare delivery. *BMJ.* 2002;325:961–964.
6. Kamaraju S, Maganty A, Davies BJ. Hospital-level variation in gastric cancer. *J Surg Oncol.* 2022;126(4):656–665.
7. OECD. Indicator 6.3: Avoidable hospital admissions. *Health at a Glance.* 2023.
8. Piñeros M, et al. Cancer patterns in Colombia. *Lancet Reg Health Am.* 2023;17:100385.
9. de Souza DLB, et al. Geographic inequalities in cancer mortality in Brazil. *PLoS One.* 2021;16(3):e0248580.
10. Hollander M, Wolfe DA. *Nonparametric Statistical Methods,* 2nd ed. Wiley; 1999.
11. White H. Heteroskedasticity-consistent covariance matrix estimator. *Econometrica.* 1980;48(4):817–838.
12. Bernal YA, et al. Multimorbidity among cancer-related hospitalizations in Chile. *Lancet Reg Health Am.* 2026;53.
13. Seabold S, Perktold J. statsmodels: Statistical modeling with Python. *Proc 9th Python in Science Conf.* 2010:92–96.
14. Pedregosa F, et al. Scikit-learn: Machine learning in Python. *J Mach Learn Res.* 2011;12:2825–2830.
15. MacKinnon JG, White H. Heteroskedasticity-consistent covariance matrix estimators. *J Econometrics.* 1985;29(3):305–325.

Submitted: May 2026 | Universidad del Desarrollo, School of Engineering | Análisis de Datos e Inferencia Estadística